

Module – 3 (Contd..)

Learning

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Learning

- Learning is the process by which a system improves its performance on a specific task by analyzing data, recognizing patterns, and adjusting its internal parameters based on the analysis.
- Learning allows AI models to make predictions, recognize patterns, and improve over time with exposure to more data.

Types of learning in AI

- **Supervised Learning:** Training is done on a labeled dataset.
Ex: Given a set of labeled images of fruits, a model can learn to classify images of apples, oranges, and bananas.
- **Unsupervised Learning:** The model is given data without explicit labels and is asked to identify patterns or groupings within the data.
Ex: Segmenting customers based on purchasing behavior.
- **Reinforcement Learning:** The model learns by interacting with its environment and receiving feedback in the form of rewards or penalties.
Ex: A model might learn to play tic-tac-toe by correcting its moves.

- **Semi-Supervised Learning:** Both labeled and unlabeled data are used to improve learning efficiency. The model is trained with a small labeled dataset and a larger set of unlabeled data to enhance its learning (useful when labeling data is time-consuming/costly).

Ex: categorizing customer feedback into positive, negative or neutral (sentiment analysis)

- **Self-Supervised Learning:** Uses data that is intrinsically structured, where part of the data can serve as labels for the other part.

Ex: In large language models, predicting a missing word in a sentence helps the model learn language structure.

1. Explanation-Based Learning

- It is a learning approach where the system learns by understanding and generalizing from a specific example by in-depth analysis.
- The knowledge so gained can be applied to new situations.

Technique

1. **Analysis of a Specific Example:** Analyze an example that performs a desired task to understand why this example works.
2. **Formulation of an Explanation:** Using logical reasoning, try to explain how the particular example works.
3. **Generalization:** From this a general rule is formulated. This generalization helps to recognize similar problems where the same principles can be applied.
4. **Application of New Knowledge:** The model uses this generalized knowledge to handle new, unseen situations, using the principles learned from the single example.

Ex: Diagnosing engine failures in cars

Suppose analysis reveals that the engine failed due to a choked fuel filter, then

- **Explaining the failure:** The system understands that the choking prevented fuel flow, leading to the engine failure.
- **Generalizing:** It learns that “any blockage in the fuel system can cause engine failure.”
- **Applying:** The model applies this understanding to other types of blockages in different car models, recognizing similar patterns of failure.

Suitability of Explanation-Based Learning

Suitable for domains where

- Some prior knowledge is readily available.
- A single, well-understood example can provide significant insights.
- Only limited data is available, but logical reasoning can still lead to effective generalization.

2. Induction learning

- It is a process where an AI model observes **patterns or relationships within data** and uses them to form general rules or hypotheses that can be applied to new, unseen data.
- It is commonly used in **supervised learning**.
- Induction learning is suitable in fields like classification, regression, and prediction, where models are trained on labeled data to make accurate predictions about new cases.

Induction Learning Process

1. **Input Examples:** The model is presented with specific examples that consist of input data along with corresponding outputs, say, training a model to recognize animals.
2. **Finding Patterns:** The model analyzes the examples to identify patterns or relationships between the input features and the output labels. (shapes, colors, dimensions, etc...of "dogs", "cats" horses, elephants,
3. **Formulation of General Rules:** From these patterns, the AI generates general rules or models. This might involve creating mathematical equations, decision trees, or other structures that represent the relationships within the data.
4. **Application to New Data:** The model uses these generalized rules to identify/predict outputs for new, unseen data.

Characteristics of Induction Learning

- **Data-Driven:** It relies heavily on data to learn. The more examples the model sees, the better it can generalize.
- **Uncertainty:** Inductive reasoning involves making guesses based on observed data. There's no guarantee that the generalization will always hold true, especially if the data is limited or biased.
- **Generalization:** Induction is fundamentally about generalization, which is crucial in allowing AI models to work well on new, unseen data.

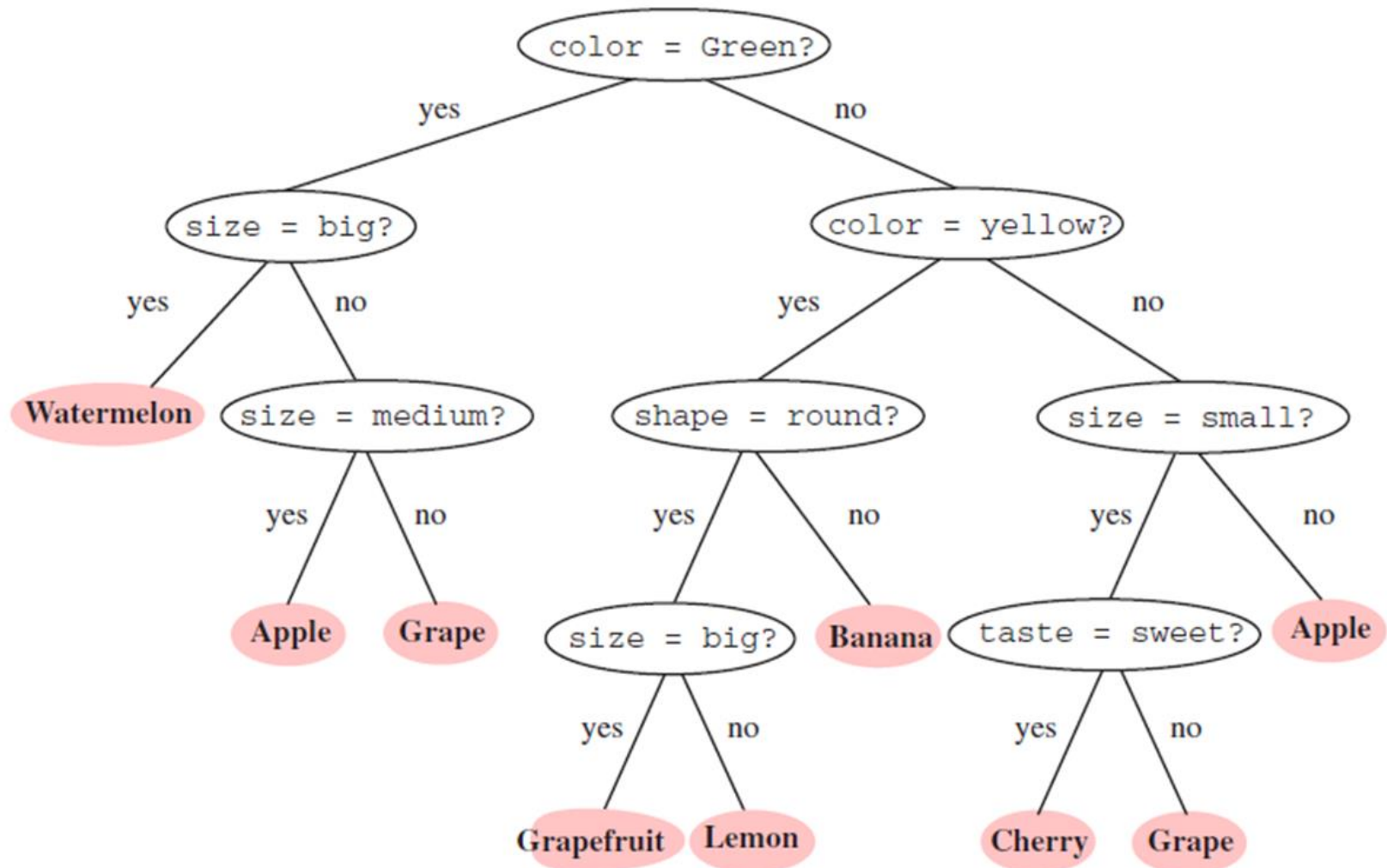
Techniques for Induction Learning

- **Decision Trees:** These models split data based on specific features, generalizing from observed patterns.
- **Support Vector Machines (SVMs):** SVMs find the best decision boundary between classes based on training data.
- **Neural Networks:** Neural networks learn by adjusting parameters based on patterns in data to generalize appropriately to new inputs.
- **K-Nearest Neighbors (KNN):** KNN classifies data by finding the closest examples in the training set.

3. Decision Tree (DT)

- It mimics human decision-making by representing decisions and their possible consequences in a tree-like structure.
- It is a popular AI model for classification and regression tasks.

Example of a Decision Tree



Structure of a DT

- **Root Node:** represents the first decision or feature that best splits the data.
- **Internal Nodes:** represent decision points based on specific features. Each node examines a feature of the data and splits it based on certain criteria (threshold value).
- **Branches:** represent the outcomes of a decision or test on a feature. For example, a branch might represent whether a value is above or below a certain threshold.
- **Leaf Nodes:** represent the final decision, e.g., in classification it may represent a class label ("spam" or "not spam"); in regression, it may represent a numeric value (salary, cost of an item, stock value).

DT Procedure

- **Splitting:** The algorithm starts at the root node and splits the dataset into subsets based on the most significant feature. It uses various criteria (Gini impurity, information gain, variance reduction) to find the best feature and threshold to split the data.
- **Recursive Partitioning:** The algorithm recursively splits each subset, creating branches and nodes until it reaches a stopping criterion (e.g., maximum depth, minimum samples per node, or when further splitting doesn't improve results).
- **Making a Decision:** Once the decision tree is built, new data can be classified or predicted by traversing the tree from the root node down to a leaf node, based on the feature values in the data.

Advantages of Decision Trees

- **Interpretability:** Decision trees are easy to understand and interpret. They provide a clear visualization of the decision-making process.
- **Can handle both Categorical and Numerical Data**

Disadvantages of Decision Trees

- **Overfitting:** Decision trees can become too complex. Pruning techniques or use of ensemble methods (e.g., random forests) help reduce overfitting.
- **Instability:** Small change in the data can lead to completely different splits, making decision trees sensitive to variations in the data.

Applications of Decision Trees

- **Classification Tasks:** customer segmentation, and medical diagnoses, spam detection
- **Regression Tasks:** Predicting house prices, stock values, or customer spending.
- **Decision-Making Systems:** Building Business rule engines, automated recommendations, and any application that requires interpretable decision-making.

4. Statistical learning

- It involves use of statistical methods to build models from data.
- It emphasizes the mathematical foundations of learning from data.
- It uses probabilistic reasoning to make predictions or understand patterns.
- This approach is particularly powerful for handling uncertainty, noise, and variability in data.

Examples of Statistical Learning Models

- **Linear and Logistic Regression:** Linear regression predicts a continuous output (predicting stock market), while logistic regression predicts categorical outcomes (whether an email is spam or not) using probability.
- **Bayesian Networks:** These probabilistic graphical models represent dependencies among variables, allowing for complex probabilistic reasoning, especially useful in decision-making tasks.
- **Naive Bayes Classifier:** Based on Bayes' theorem.

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- **Hidden Markov Models (HMMs):** HMMs are used for **time-series data** where the model predicts a sequence of states based on observed variables. It's widely used in applications like speech recognition and natural language processing.
- **Support Vector Machines (SVMs):** Used to find decision boundaries to classify data by maximizing the margin between classes.
- **Principal Component Analysis (PCA):** It reduces the number of features while preserving variance (feature selection). Thus, helps in reducing computation.

Limitations of Statistical Learning

- **Requires Large Datasets:** Statistical learning often needs large amounts of data to accurately estimate parameters and make reliable predictions.
- **Complexity:** The mathematical basis can make it challenging to implement and interpret, especially for non-experts.
- **Assumptions about Data:** Many statistical models rely on assumptions about data distribution; if these assumptions are incorrect, model performance may suffer.

Applications

- **Natural Language Processing (NLP):** Used in text classification, sentiment analysis, and language modeling.
- **Computer Vision:** Statistical models are applied for tasks like object detection and face recognition.
- **Finance:** Stock price prediction, risk assessment, and applications that require handling of uncertainty.
- **Healthcare:** Used for diagnosing diseases, predicting patient condition, and understanding patterns in medical data.

5. Bayesian learning

- It is based on Bayes' theorem:
$$P(H|E) = \frac{P(E|H) \times P(H)}{P(E)}$$
- **P(H|E) is the posterior probability:** the probability of the hypothesis H given the evidence E.
- **P(E|H) is the likelihood:** the probability of observing the evidence E given that the hypothesis H is true.
- **P(H) is the prior probability:** the initial belief about the probability of the hypothesis before seeing the evidence.
- **P(E) is the evidence:** the overall probability of observing E, regardless of H.

In Bayesian learning, the model updates its beliefs (posterior) about a hypothesis each time it receives new evidence, combining both prior knowledge and the observed data.

Bayesian Learning Models

- Naive Bayes Classifier
- Bayesian Networks
- Bayesian Linear Regression
- Gaussian Mixture Models (GMMs)
- Hidden Markov Models (HMMs)
- Markov Chain Monte Carlo (MCMC)

Applications of Bayesian Learning

- Medical Diagnosis
- Natural Language Processing
- Robotics and Control Systems
- Recommendation Systems

Bayesian methods are useful in domains where data is limited, interpretability is crucial, or decisions are to be made under uncertainty.

Expectation Maximization (EM)

- It is an iterative optimization algorithm commonly used in machine learning and statistics to find the **maximum likelihood estimates** of parameters in models with **latent (hidden) variables**.
- It can handle missing values and estimate parameters in the presence of unobserved variables. Thus, useful when working with **incomplete or uncertain data**.

Technique

The EM algorithm alternates between two main steps: the **Expectation (E) step** and the **Maximization (M) step**. The algorithm repeats these steps iteratively until convergence (when changes in the parameter estimates are very small).

1. **Initialization:** Start with initial guesses for the parameters. The initial choice can affect the convergence and speed.
2. **Expectation (E) Step:** In this step, the algorithm calculates the expected value of the latent variables (hidden or unobserved variables) given the observed data and the current estimates of the parameters.
3. **Maximization (M) Step:** Using the expected values obtained in the E step, the algorithm maximizes the **likelihood function** to update the parameter estimates.
4. **Iteration:** The algorithm alternates between the E and M steps until it converges.

Applications

- **Clustering:** EM is used in clustering algorithms like Gaussian Mixture Models, where it identifies clusters based on the distribution of data points.
- **Computer Vision:** In image processing, EM can help in segmentation tasks by modeling pixel intensities as mixtures of Gaussians.
- **Genetics and Bioinformatics:** EM is used in applications where the latent variables represent unknown genetic factors.
- **Speech Recognition:** In speech models, EM helps in estimating the parameters of Hidden Markov Models (HMMs), where the underlying phoneme sequence is treated as a latent variable.

Hidden Markov model(HMM)

- A hidden Markov Model relies on the assumption that the events we observe depend on some internal factors or states that are not directly observable.
- It is a **statistical model** used to represent systems that follow a Markov process with unobserved (hidden) states.
- Suitable for situations where one has to **infer a sequence of hidden (latent) states based on observed data**.
- **Markov process:** A Markov process is a type of stochastic (random) process that satisfies the Markov property, which states that the **future state of the process depends only on the current state and not on any previous states**.

Characteristics of a Markov Process

1. **Memoryless:** The probability of transitioning to the next state depends only on the current state. Hence it is not required to store the earlier states.
2. **Discrete States:** Markov processes often have a discrete set of possible states.
3. **Transition Probabilities:** Each state has a set of probabilities for transitioning to other states.

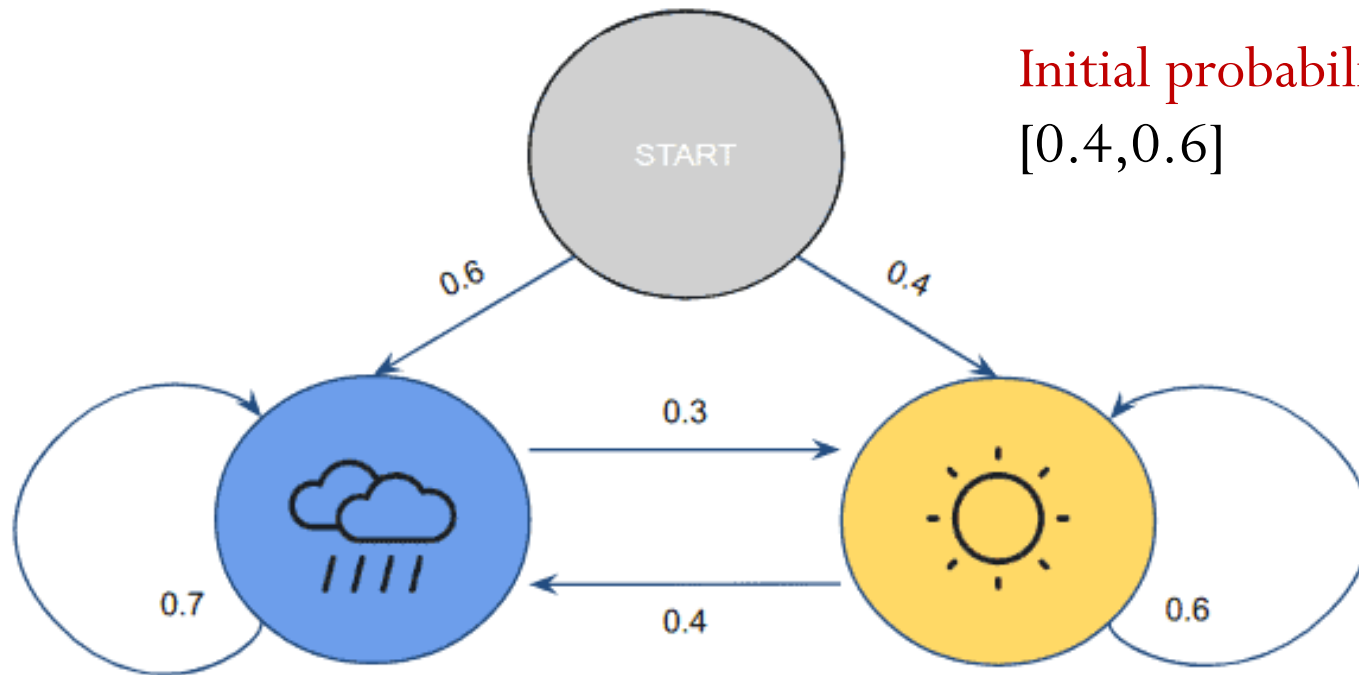
Components of HMM

- **Hidden States:** The underlying states of the system that are not directly observable. For example, in speech recognition, the hidden states might represent phonemes.
- **Observed States:** These are the visible outputs or observations generated by the hidden states. In speech recognition, these could be sound wave features.

Ex: Guessing game (weather prediction based on one's activity)

- Anil and Arun are close friends who live far away from each other, but they talk over phone every day and discuss what they do. They decide to play a guessing game in which Anil tries to guess the weather based only on what Arun tells him, he is doing that day.
- Let's imagine Arun's behavior is limited. He can do one of three things during the day – walk in the park, go for shopping, or clean his apartment. Additionally, his actions fully depend on the weather of the given day - rainy or sunny.

Initial Guess



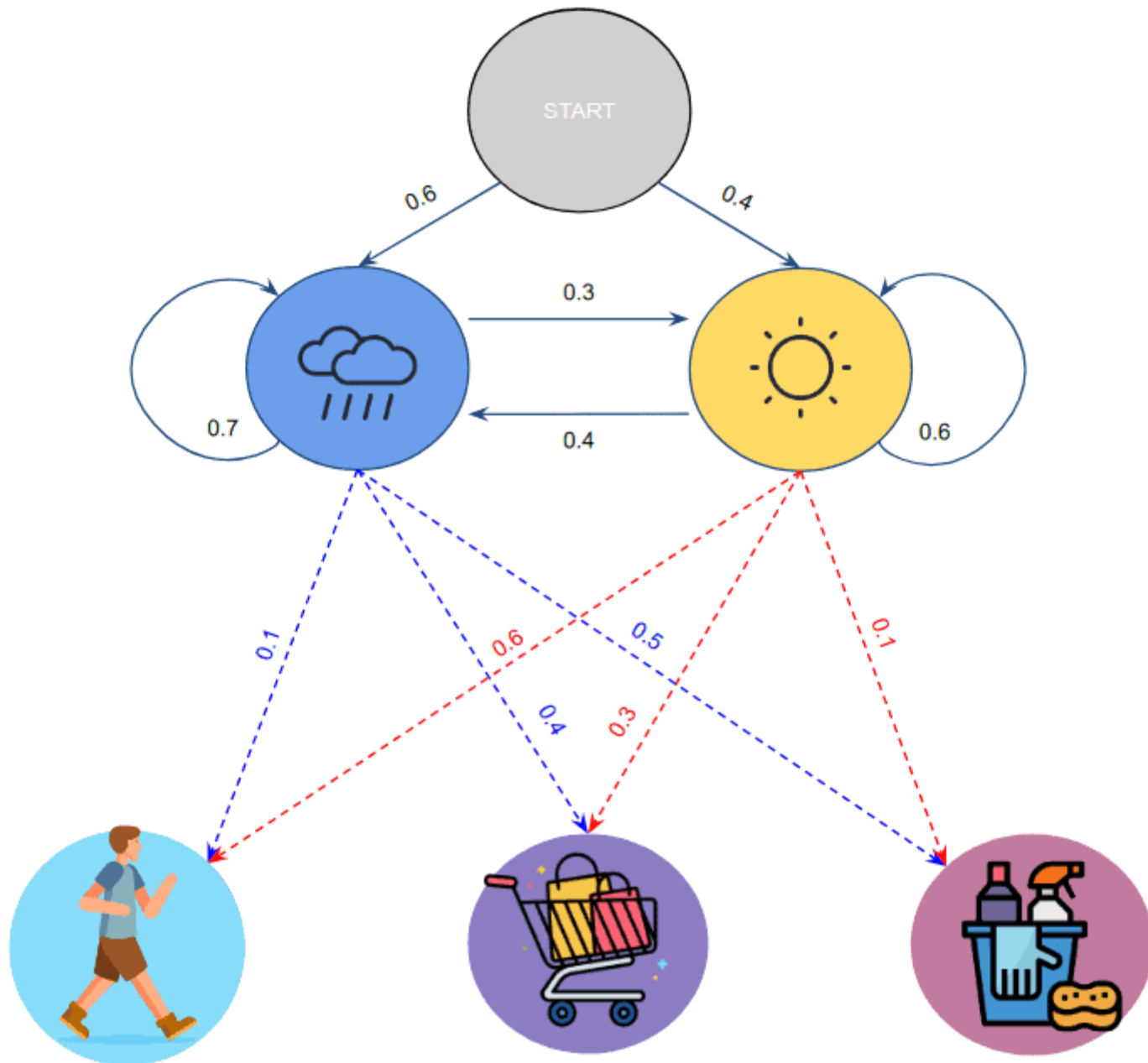
Initial probabilities:
[0.4,0.6]

If it was raining yesterday, there is a 0.7 probability that it will rain again and a 0.3 probability that it would be sunny.

If yesterday was sunny, there is a 0.6 probability that today will be sunny again and 0.4 probability that it'll rain

Transition Probabilities

- Since the activities Arun will do depend fully on the weather, we can assign **conditional probabilities** for each state, which are known as **emission probabilities**.
- For example, if today is sunny, there is a 0.6 probability that Arun will go for a walk, 0.3 for shopping, and 0.1 for cleaning his house. Since Anil talks with him each day, he has gathered a sequence of observations of his actions.



Parameters in an HMM

1. **Transition Probabilities (T):** This is a matrix representing the probabilities of transitioning from one hidden state to another. Each entry T_{ij} represents the probability of transitioning from hidden state i to hidden state j .
2. **Emission Probabilities (E):** This matrix describes the probability of an observed state given a hidden state. Each entry E_{jk} represents the probability of observing k given that the system is in hidden state j .
3. **Initial State Probabilities (I):** These probabilities represent the likelihood of starting in each hidden state.

Important Steps

- **Evaluation:** Given a sequence of observations, calculate the probability that this sequence was generated by the HMM. This is useful for tasks like speech recognition, where we want to determine if a sound matches a known pattern.
- **Decoding:** Find the most likely sequence of hidden states given a sequence of observations. For example, in part-of-speech tagging, this means identifying the sequence of tags (e.g., noun, verb) given a sentence.
- **Learning (Parameter Estimation):** Given observed sequences, determine the best parameters (transition probabilities, emission probabilities, and initial probabilities) that describe the HMM.

Applications of HMM

- **Speech Recognition:** Used to model phonemes and words as sequences of hidden states based on acoustic observations.
- **Natural Language Processing:** where the next word in a sentence depends on the previous word.
- **Bioinformatics:** Used to analyze DNA sequences, protein structures, and other biological data by modeling sequences of genes or amino acids.
- **Finance:** Applied in modeling stock price trends where hidden states represent market conditions.
- **Weather Prediction:** where the next day's weather depends on the current day.

Closed world problems

- **Closed-world problems** refer to scenarios where it is assumed that all relevant information about the environment or domain is completely known.
- This assumption allows the AI system to operate with the belief that if a fact or entity is not explicitly stated or known, it can be considered false or non-existent.

Closed World vs. Open World Assumptions

- **Closed-World Assumption (CWA):** Assumes that if something is **not known to be true, it is false**. This is **suitable for environments where all information is known and well-defined**.
- **Open-World Assumption (OWA):** Assumes that lack of information does not imply falsehood; it only means that the system doesn't know. This assumption is more **suitable for complex, dynamic, and uncertain environments where new information may arise at any time**.

Examples of Closed-World Problems

- **Database Queries:** If a query does not return certain data, the system assumes that the data does not exist within the database.
- **Game Playing:** Games like chess operate under closed-world assumptions, as the rules, moves, and possible states are fully known and defined.
- **Expert Systems:** Traditional expert systems, such as medical diagnosis tools or rule-based systems, often operate in a closed world, where the rules and symptoms for diagnosing conditions are predefined, and the system does not account for unknown diseases or symptoms.